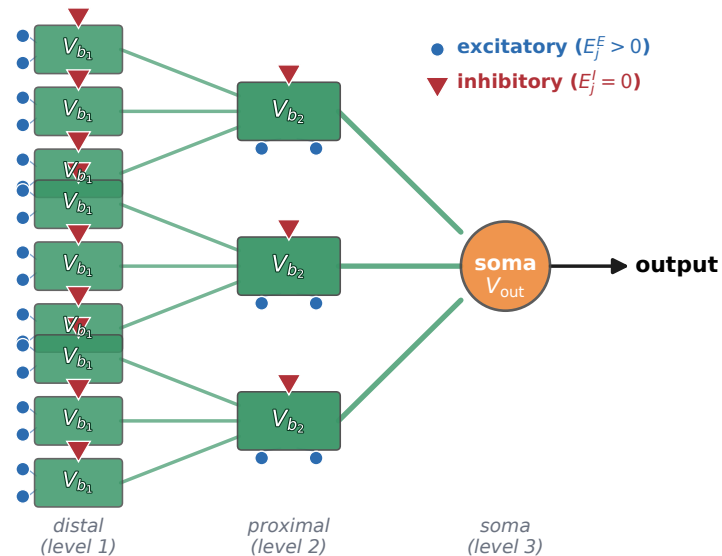


A Compartmental dendritic neuron: shunting E/I integration



B Shunting voltage integration

Each branch aggregates its synapses via a conductance equation:

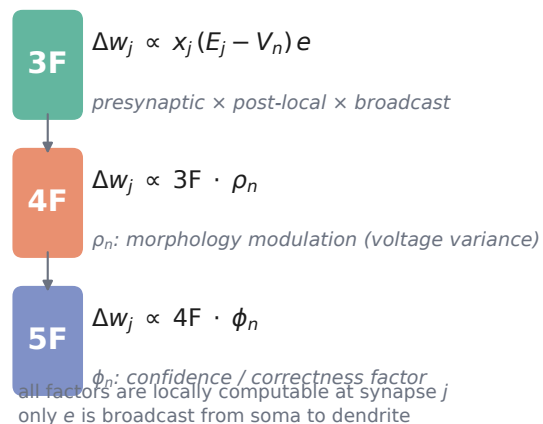
$$V_n = \frac{\sum_j g_j x_j E_j}{g_n^{\text{tot}}}$$

$$g_n^{\text{tot}} = g^{\text{leak}} + \sum_{j \in E} g_j x_j + \sum_{j \in I} g_j x_j$$

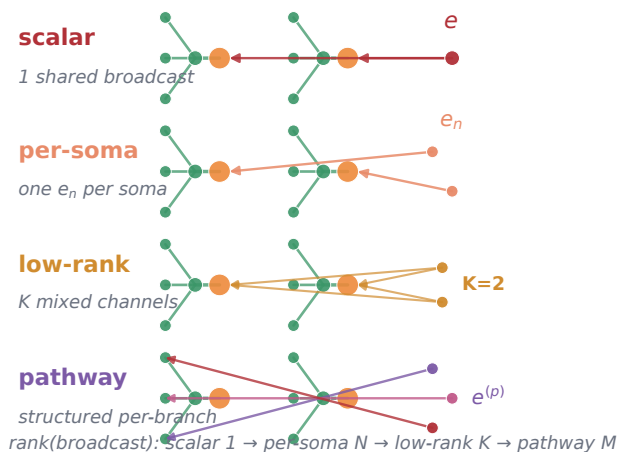
shunting: inhibition raises g_n^{tot} and lowers gain
 additive broad gain dist.



C Local rule family (3F → 4F → 5F)



D Broadcast channels for e



E MNIST learning dynamics ([3, 3] shunting)

